Aneesh Ganti

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EDUCATION

University of Illinois Urbana-Champaign Expected: May 2027

Bachelor of Science in Computer Science and Mathematics, Minor in Electrical Engineering *GPA: 3.75/4.00*

Relevant Coursework: • Data Structures • Algorithms • Distributed Systems • Cloud Computing • Database Systems • Software Design • Applied Machine Learning • Deep Learning • Computer Vision • System Programming • Computer Architecture • Numerical Method

PROFESSIONAL EXPERIENCE

Machine Learning Engineer Intern | Brunswick Corporation January 2026 — Present

- Engineered propeller design tool processing **1,000+** boat configurations to optimize its speed, fuel economy, and handling metrics.
- Trained PyTorch models to predict propeller design impacts with **92%** accuracy, reducing test costs by **$3,000 per configuration**.
- Deployed FastAPI backend and React UI for Mercury's commercial platform to deliver real-time recommendations for **500+ users**.

Software Engineer Intern | Tekweld Manufacturing May 2025 — Feb 2026

- Developed sales automation tool for targeted outreach campaigns that generated **90,000+** client leads with **93%** contact success.
- Accelerated product design process from **1 week to 15 minutes** by creating an LLM-driven pipeline for design and quality control.
- Implemented UI and backend with TypeScript, React, CSS, Python, PostgreSQL, RESTful API endpoints, and Amazon Web Services.

Software Engineer Intern | PhinD Experts May 2025 — August 2025

- Built Python-based web scraping platform to extract **4,000+** PhD profiles from Google Scholar for recruiter-candidate database.
- Revamped ETL pipeline with SQLAlchemy and Airflow to reduce processing time for user matching from **1 minute to 25 seconds.**
- Automated cloud infrastructure provisions with GitHub Actions, Docker, and AWS to support scaling and new feature deployment.

RESEARCH EXPERIENCE

Undergraduate Researcher | University of Illinois Urbana-Champaign February 2026 — Present

- Developed 3D motion forecasting system of objects in RGB images to improve robotic perception and manipulation performance.
- Integrated lifting model with PyTorch to impute 3D pseudo-labels from 2D annotations, increasing labeled training data by **260%**.
- Architected a ViT-conditioned diffusion forecasting model to predict multimodal object keypoint motion over **17-second** horizons.

Machine Learning Researcher | University of California, Santa Barbara June 2023 — August 2023

- Implemented deep neural network with TensorFlow and Keras to predict degradation of lithium-ion batteries in electric vehicles.
- Reduced battery life prediction error by **18%** to outperform baseline models using NASA's dataset of **10,000+** charging cycles.
- Collaborated with UCSB and MIT AI researchers to develop and present findings at capstone seminar for advancing EV diagnostics.

LEADERSHIP AND ACTIVITIES

Course Assistant | Siebel School of Computing and Data Science January 2025 — Present

- Host CS 173 Discrete Structures office hours for **700+** students, covering formal logic, proof writing, graph theory, and algorithms.
- Conduct weekly exam review sessions for **250+** students, boosting their exam performance worth **90%** of course grade by **15%**.

Software Engineer | Spaceshot Rocketry Program August 2024 — Present

- Implemented C++ Kalman Filter to estimate rocket dynamics with **10-ft** accuracy in **75,000-ft** launch for precise flight control.
- Built telemetry monitoring interface with JavaScript, HTML, and CSS to process MQTT packet transmissions during live launches.
- Optimized autonomous turret control software to ensure zero telemetry data packet loss throughout complete flight trajectory.

Software & Simulations Lead | NASA Human Lander Challenge August 2024 — July 2025

- Engineered ML pipeline using Python, ANSYS, and Scikit-learn to improve in-space cryogen propellant transfer efficiency by **84%**.
- Designed agnostic transfer protocol evaluated by NASA's Cryogenic Fluid Management team for Artemis program implementation.
- Led team technical presentation to win Best Presentation and **$1,200** amongst **30+** universities at Marshal Space Flight Center.

SKILLS

Programming Languages: Python, C++, TypeScript, JavaScript, SQL, Java, HTML, CSS, Swift, Verilog, MIPS, R
Full-Stack & Backend: React, NodeJS, Express, FastAPI, Flask, Django, WebSockets, REST APIs, PostgreSQL, MongoDB, Redis
ML & Data Science: PyTorch, TensorFlow, Scikit-learn, Keras, LangChain, OpenCV, Hugging Face, Gemini, NumPy, Pandas
Other: Git, GitHub, CI/CD, GitHub Actions, Docker, Kubernetes, Amazon Web Services, Google Cloud, Agile, Apache, Linux, Jest